# Technical Summary Sheet 6.5930 / Hardware Architectures for Deep Learning, Spring 2026

**Topology-Aware Mapping for Photonic Interconnects in TPU-Class LLM Accelerators**

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## Repository

`final-project/workspace/final_project/` (extends the AccelForge cluster example). Reproduce all numbers with `make study && make notebook && make test` from `final-project/`. The 71-test suite is the canonical correctness check; nine figures live under `results/presentation/`.

## Three Key Insights / Innovations

1. **The intra-chip mapping is stable; the distributed schedule is what changes.** For the 64-chip TPU cube, AccelForge converges to the same partitioning/dataflow/tile signature for all seven technologies (**0/24** inner mappings change). The benefit lives one layer up: **5/16** workload–scenario pairs flip their optimal communication pattern, and the optimal compiler-visible schedule changes whenever multicast is available. Separating the two layers matches the 6.5930 mapping formalism rather than conflating it.
2. **Hardware multicast, not raw bandwidth, dominates shared-state workloads.** A 3.5 Tb/s, 6 pJ/bit advanced chiplet baseline—chosen to defeat the bandwidth hypothesis—is still beaten by M1000 by **99.85%** on shared-DRAM multicast: $N$ unicasts collapse to one broadcast, and contention disappears on a switched fabric. The same chiplet is competitive on node-to-node exchange.
3. **Photonics is not a uniform multiplier.** On a 9216-chip Ironwood-class superpod, long-context inference reaches **14.50**× speedup, but training only reaches **1.43**×. Once collectives shrink, training becomes compute-dominated; inference stays network-exposed because decode and MoE routing cross the fabric every step. The strong case for photonics is inference and long-context serving, not training payoff.

## Headline Results (reproducible from `results/`)

| Axis | Value | Setting |
|---|---|---|
| 16-node energy reduction | 72.21% | EVK100 multicast |
| 16-node latency reduction | 97.57% | EVK100 multicast |
| TPU-cube network energy red. | 99.91% | M1000 multicast |
| TPU-cube AccelForge energy red. | 77.70% | M1000 KV attention |
| Inner mapping signatures | 0/24 changed | TPU cube |
| Distributed maps changed | 5/16 cases | cluster study |
| Compiler-schedule best gain | 67.75% | M1000 remote streaming |
| Multicast scaling exponent | 0.00 | M1000 vs torus / chiplet |
| Chiplet-relative reduction | 99.85% | M1000 multicast |
| Superpod inference speedup | 14.50× | Deep Think proxy |
| Superpod training speedup | 1.43× | Pro MoE proxy |
| Superpod inference energy red. | 51.07% | Deep Think proxy |

## Contributions

**Simone Surdi** – AccelForge/network model: topology classes (ring, mesh, 3-D torus, optical crossbar / switched fabric); the `LogicalClusterMemory` extension; the iterative network-aware feedback loop (`solve_iteratively`); mapping-signature extraction; cluster, TPU-cube, and chiplet-baseline studies; technology-catalog constants and sensitivity sweeps.
**Pranav Sitaraman** – Compiler/superpod analysis: schedule enumeration (SPMD collectives, remote HBM streaming, replicated HBM cache, multicast cache fill, MoE locality, full photonic stack); the 9216-chip Ironwood-class stage model (prefill / decode / refill / MoE / training / gradient / optimizer); Gemini-class proxy workloads; ablation ladder; integration tests; report, poster, and presentation deliverables.

## Reproducibility and Threats to Bias

Lightmatter constants are from public product pages and SC25/ISES talks; TPU constants are from Cloud TPU docs and the Ironwood blog / Hot Chips tutorial. EVK100 is reported both as the conservative 3.2 pJ/bit product target and the 1.9 pJ/bit Passage-family sensitivity point; M1000 uses 114.6 Tb/s bidirectional and 2.3 pJ/bit including laser power. Optical multicast is modeled as one logical broadcast (optimistic but explicit). The superpod stage model is analytical with an overlap factor; cycle-accurate simulation is left to future work. Gemini parameter counts are unpublished, so we use public proxy stress workloads. The 71-test suite asserts technology constants, the multicast scaling exponent, and the schedule-changes-under-photonics property as regression tests.